

Project Report: Automated API + UI Test Suite - QA & Test Engineer

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Family:	QA & Test Engineer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

pytest API tests + Playwright E2E for login flow; run in GitHub Actions on every PR.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview** *(Level: Intermediate)

Automated API + UI Test Suite** is a intermediate-level end-to-end project for QA & Test Engineer. **Duration:** 3-4 weeks. **pytest API tests + Playwright E2E** for login flow; run in GitHub Actions on every PR. **Tools:** pytest, Playwright, GitHub Actions, httpx.

Code Reference:

```
Objective: pytest API test collection
Objective: Playwright E2E login + navigation
Objective: GitHub Actions CI pipeline
Objective: Test coverage report for API
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

QA & Test Engineer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a QA & Test Engineer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: API [Intermediate]

Chapter focus: Phase: API** *(Level: Intermediate)

pytest + httpx tests. Complete these tasks: Auth endpoints; CRUD tests. Deliverable for this phase: API test suite.

Code Reference:

- [] Auth endpoints
- [] CRUD tests

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase API is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the API test suite to the teacher.

Chapter 3: Phase: E2E [Intermediate]

Chapter focus: Phase: E2E** *(Level: Intermediate)

Playwright browser tests. Complete these tasks: Login flow; Page assertions. Deliverable for this phase: E2E video recordings.

Code Reference:

- [] Login flow
- [] Page assertions

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase E2E is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the E2E video recordings to the teacher.

Chapter 4: Phase: CI [Intermediate]**Chapter focus: Phase: CI** *(Level: Intermediate)**

GitHub Actions workflow. Complete these tasks: Run on PR; Fail on red. Deliverable for this phase: CI badge green.

Code Reference:

- [] Run on PR
- [] Fail on red

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase CI is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the CI badge green to the teacher.

Chapter 5: Phase: Report [Intermediate]**Chapter focus: Phase: Report** *(Level: Intermediate)**

Allure or HTML report. Complete these tasks: Trends; Flaky test tracking. Deliverable for this phase: Test dashboard HTML.

Code Reference:

- [] Trends
- [] Flaky test tracking

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Report is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

QA & Test Engineer agile sprints and coursework milestones.

Real use example

Team demo shows the Test dashboard HTML to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review *(Level: Intermediate)**

Submit all final deliverables: Test repo; CI pipeline; HTML report; E2E videos. Skills demonstrated: pytest, Playwright, CI/CD. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Test repo
Deliverable: CI pipeline
Deliverable: HTML report
Deliverable: E2E videos
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide

What it actually is

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Test repo and passes the rubric.